

SQL Challenges

Set up the Classic Models dataset locally with example data using the SQL script: <https://raw.githubusercontent.com/harsha547/ClassicModels-Database-Queries/master/database.sql>

1. Prepare a list of offices sorted by country, state, city.

???

2. How many employees are there in the company?

???

3. What is the total of payments received?

???

4. List the product lines that contain 'Cars'.

???

5. Report total payments for October 28, 2004.

???

6. Report those payments greater than \ \$100,000.

???

7. List the products in each product line.

???

8. How many products in each product line?

???

9. What is the minimum payment received?

???

10. List all payments greater than twice the average payment.

???

11. What is the average percentage markup of the MSRP on buyPrice?

???

12. How many distinct products does ClassicModels sell?

???

13. Report the name and city of customers who don't have sales representatives?

???

14. What are the names of executives with VP or Manager in their title? Use the CONCAT function to combine the employee's first name and last name into a single field for reporting.

???

15. Which orders have a value greater than \$5,000?

???

Solution

Summary and Further Reading

The following topics were covered in this tutorial:

- Use cases and design of relational databases and SQL
- Installation of MySQL and setting up a database locally
- Creating, modifying and deleting databases and database tables
- SQL Data types and constraints (primary key, foreign key)
- CRUD (Create, Read, Update and Delete) operations on tables
- Exporting and importing data from relational databases

Check out the following resources to learn more:

- ClassicModels Database: <https://www.mysqltutorial.org/mysql-sample-database.aspx>
- SQL Tutorial: <https://www.w3schools.com/sql/default.Asp>
- SQL Books & Slides: <https://www.db-book.com/slides-dir/index.html>
- SQL Exercises: <https://www.w3resource.com/sql-exercises/>
- MySQL Reference Manual: <https://dev.mysql.com/doc/refman/8.0/en/>